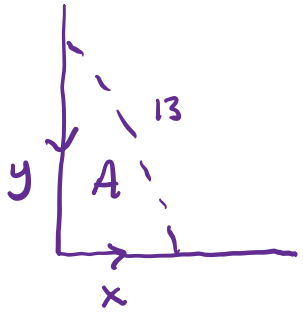
Problem 15. TIER 4 — SAME PICTURE, DIFFERENT QUESTION

A 13 ft ladder leans against a wall, and its base slides away from the wall at 2 ft/s. How fast is the **area of the triangle** formed by the ladder, the wall and the ground changing when the base is 5 ft from the wall?

① 

② $\frac{dx}{dt} = 2\ ft,\quad \frac{dA}{dt} = ?\quad$ when $x = 5$

③ $A = \frac{1}{2}(b)(h) = \frac{1}{2}xy$

$x^2 + y^2 = 13^2$

④ $\frac{d}{dt}A = \frac{d}{dt}\frac{1}{2}xy$

$\frac{dA}{dt} = \frac{1}{2}\left[x\frac{dy}{dt} + y\frac{dx}{dt}\right]$

$\frac{d}{dt}(x^2 + y^2) = \frac{d}{dt}13^2$

$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$

$\frac{dy}{dt} = \frac{-x}{y}\frac{dx}{dt}$

get $\frac{dy}{dt}$

get y:



$5^2 + y^2 = 13^2$

$25 + y^2 = 169$

$y^2 = 169 - 25 = 144$

$y = 12$

$\frac{dy}{dt} = \frac{-5}{12}2 = -\frac{5}{6}$

get $\frac{dA}{dt} = \frac{1}{2}\left[5\cdot\left(-\frac{5}{6}\right) + 12\cdot 2\right]$

$= \frac{1}{2}\left[\frac{-25}{6} + \frac{24\cdot 6}{6}\right] = \frac{1}{2}\left[\frac{-25+144}{6}\right] = \frac{119}{12}\ \frac{ft^2}{sec}$

$\approx 9.92\ \frac{ft^2}{sec}$